

Crime Analysis – Tableau Project

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1. Project Overview

This project analyzes historical crime data using Tableau to identify crime patterns and trends that can support law enforcement planning and crime prevention strategies. The analysis covers the geographical distribution of incidents, common crime types, time-based patterns, year-over-year trends, arrests, and incident severity. Four interactive dashboards were developed to transform the crime data into clear and actionable insights.

2. Dashboard 1 — Overall Crime Statistics Dashboard

2.1. Crime Across Districts — Bubble Map: Crime counts vary across different districts with no definite pattern.

2.2. Common Criminal Incidents — Bar Chart: Theft has the highest number of reported incidents at 58,434, while Burglary has the lowest among the top eight criminal incidents with more than 10,000 reported occurrences, at 11,053.

2.3. Recent Crimes in the Current Year — Table: Criminal Trespass, Criminal Damage, Other Offense, Theft, and Robbery were reported in the latest year, 2022, with each displayed incident occurring once.

3. Dashboard 2 — Time Period Analysis Dashboard

3.1. Incidents Reported by Week — Line Chart: Monday recorded the highest number of incidents at 36,286, while Friday recorded the lowest at 34,096.

3.2. Incidents Reported by Time Blocks — Bar Chart: Afternoon had the highest number of reported incidents at 67,001, while Night had the lowest at 33,361.

3.3. Incidents Reported by the Hour — Line Chart: Incidents peaked at 12:00 PM with 15,562 incidents, while the lowest count was recorded at 5:00 AM with 3,686 incidents.

4. Dashboard 3 — Trend Analysis Dashboard

4.1. Incidents Reported on a Specific Date — Line Chart: Shows the trend in incidents reported on a selected date across different years, along with the year-over-year change.

4.2. Crime Rate Over the Years — Table: Shows the number of incidents reported for each year and the year-over-year absolute & percentage change. Reported incidents were highest in 2017 at 102,602 and lowest in 2022 at 6 incidents.

4.3. Incidents Reported on a Specific Date — Table: Shows the number of incidents reported on the selected date for each year, along with the year-over-year change in the number of incidents.

5. Dashboard 4 — Comparative Analysis Dashboard

5.1. Arrest Made Vs Not Made — Pie Chart: Out of the reported incidents, 20.28% (50,253) resulted in an arrest, while 79.72% (197,544) did not result in an arrest.

5.2. Incident Severity — Bar Chart: Shows different crime/incident types and their descriptions, along with the severity percentage associated with each description.

6. Conclusion

This project provides a comprehensive view of the crime data through interactive Tableau dashboards. The findings offer a foundation for further analysis and can support more informed, data-driven approaches to understanding and monitoring crime patterns.